

W04 - Software Installation on HPC

Mohammad Amin GHAVAMI

Uni.lu HPC username: "mghavami"

GitHub username: "AminGhavami"

University of Luxembourg

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1 Question 2: Inspecting the SCOTCH modulefile

Answer all the questions below in your report:

- What is the full path to the SCOTCH modulefile?
`/opt/apps/resif/aion/2020b/epyc/modules/all/math/SCOTCH/6.1.0-gompi-2020b`
- What is the homepage of the SCOTCH library, as given in the module?
`https://gforge.inria.fr/projects/scotch/`
- What is the installation path of the SCOTCH library?
`/opt/apps/resif/aion/2020b/epyc/software/SCOTCH/`

This version of SCOTCH has been installed with EasyBuild.

- Can you tell which version of EasyBuild has been used to install it? Where did you find this information?
By running `cat /opt/apps/resif/aion/2020b/epyc/modules/all/math/SCOTCH/6.1.0-gompi-2020b.lua`
: Built with EasyBuild version 4.4.1
- Can tell where all the modulefiles are installed? `/opt/apps/resif/aion/2020b/epyc/modules/all`
- How many modulefiles are installed in this path? You can use `find PATH -type f -name '*.lua' | wc -l` to count all the .lua files in a given path.

```
0 [mghavami@aion-0185 ~](6948342 1N/1T/1CN)$ find /opt/apps/resif/aion/2020b/epyc/modules/all -type f -name '*.lua' | wc -l
324
```

Figure 1: Answers to the question

2 Question 3: Modulefile for preCICE installation

Add your Modulefile for the preCICE installation to your GitHub Classroom repository, at the location defined above.

Done!

3 Question 4: Testing the preCICE modulefile

- Run the commands above to test your preCICE modulefile

```

0 [mghavami@aion-0104 y2425-w04-software-installation-on-hpc-AminGhavami] (6910572 1N/1T/1CN)$ which precice-tools
~/precice-install/bin/precice-tools
0 [mghavami@aion-0104 y2425-w04-software-installation-on-hpc-AminGhavami] (6910572 1N/1T/1CN)$ precice-tools --version
3.1.2;no-info [git failed to run];PRECICE_FEATURE_MPI_COMMUNICATION=Y;PRECICE_FEATURE_PETSC_MAPPING=Y;PRECICE_FEATURE_GINKGO_MAPPING=N;PRECICE_FEATURE_PYTHON_ACTI
ONS=Y;PRECICE_BINDINGS_C=Y;PRECICE_BINDINGS_FORTTRAN=Y;CXX=GNU;CXXFLAGS=-march=native -O3 -O3 -DNDEBUG;LDFLAGS=
0 [mghavami@aion-0104 y2425-w04-software-installation-on-hpc-AminGhavami] (6910572 1N/1T/1CN)$ cd precice/dummy_solver_cpp/
0 [mghavami@aion-0104 dummy_solver_cpp] (6910572 1N/1T/1CN)$ cmake .
-- The CXX compiler identification is GNU 10.2.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /opt/apps/resif/aion/2020b/epyc/software/GCCcore/10.2.0/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Looking for C++ include pthread.h
-- Looking for C++ include pthread.h - found
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Failed
-- Looking for pthread_create in pthreads
-- Looking for pthread_create in pthreads - not found
-- Looking for pthread_create in pthread
-- Looking for pthread_create in pthread - found
-- Found Threads: TRUE
-- Configuring done
-- Generating done
-- Build files have been written to: /home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/dummy_solver_cpp

```

Figure 2: Answers to the question

- Take a screenshot of your terminal output and Add the screenshot to your report

```

0 [mghavami@aion-0104 dummy_solver_cpp] (6910572 1N/1T/1CN)$ make
Scanning dependencies of target solverdummy
[ 50%] Building CXX object CMakeFiles/solverdummy.dir/solverdummy.cpp.o
[100%] Linking CXX executable solverdummy
[100%] Built target solverdummy
0 [mghavami@aion-0104 dummy_solver_cpp] (6910572 1N/1T/1CN)$ ctest -V
UpdateCTestConfiguration from :/home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/dummy_solver_cpp/DartConfiguration.tcl
UpdateCTestConfiguration from :/home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/dummy_solver_cpp/DartConfiguration.tcl
Test project /home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/dummy_solver_cpp
Constructing a list of tests
Done constructing a list of tests
Updating test list for fixtures
Added 0 tests to meet fixture requirements
Checking test dependency graph...
Checking test dependency graph end
test 1
Start 1: precice.solverdummy

1: Test command: /mnt/aiongpf/apps/resif/aion/2020b/epyc/software/CMake/3.18.4-GCCcore-10.2.0/bin/cmake "--DDUMMY_EXE=/home/users/mghavami/y2425-w04-software-inst
allation-on-hpc-AminGhavami/precice/dummy_solver_cpp/solverdummy" "--DDUMMY_CONFIG=/home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/
dummy_solver_cpp/precice-config.xml" "-V" "-P" "/home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/dummy_solver_cpp/test.cmake"
1: Test timeout computed to be: 10000000
1: Running Two Solverdummies
1: Using executable: /home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/dummy_solver_cpp/solverdummy
1: Using config: /home/users/mghavami/y2425-w04-software-installation-on-hpc-AminGhavami/precice/dummy_solver_cpp/precice-config.xml
1: The solverdummy was called with an incorrect number of arguments. Usage: ./solverdummy configFile solverName
1:
1: Parameter description
1: configurationFile: Path and filename of preCICE configuration
1: solverName: SolverDummy participant name in preCICE configuration
1: Success!
1/1 Test #1: precice.solverdummy ..... Passed 0.05 sec

100% tests passed, 0 tests failed out of 1

Total Test time (real) = 0.05 sec

```

Figure 3: Answers to the question

4 Question 5: EasyBuild initial setup

Answer to these questions in your report:

- Take a screenshot of the 3 steps above and add it to your report.

```

1 [mghavami@aion-0104 eessi](6910572 1N/1T/1CN)$ module avail EasyBuild
---- /cvmfs/software.eessi.io/versions/2023.06/software/linux/x86_64/amd/zen2/m
odules/all ----
  EESSI-extend/2023.06-easybuild      EasyBuild/4.9.1      EasyBuild/4.9.4
  EasyBuild/4.8.2                     EasyBuild/4.9.2      EasyBuild/5.0.0 (D)
  EasyBuild/4.9.0                     EasyBuild/4.9.3
----- /opt/apps/resif/aion/2020b/epyc/modules/all -----
  tools/EasyBuild/4.4.1      tools/EasyBuild/4.5.4 (D)

Where:
  D: Default Module

If the avail list is too long consider trying:

"module --default avail" or "ml -d av" to just list the default modules.
"module overview" or "ml ov" to display the number of modules for each name.

Use "module spider" to find all possible modules and extensions.
Use "module keyword key1 key2 ..." to search for all possible modules matching
any of the "keys".

0 [mghavami@aion-0104 eessi](6910572 1N/1T/1CN)$ module load tools/EasyBuild/4.5.4
0 [mghavami@aion-0104 eessi](6910572 1N/1T/1CN)$ eb --version
This is EasyBuild 4.5.4 (framework: 4.5.4, easyblocks: 4.5.4) on host aion-0104.

0 [mghavami@aion-0104 eessi](6910572 1N/1T/1CN)$ eb --show-config
#
# Current EasyBuild configuration
# (C: command line argument, D: default value, E: environment variable, F: configuration file)
#
buildpath          (E) = /tmp/mghavami
containerpath      (D) = /home/users/mghavami/.local/easybuild/containers
installpath        (D) = /home/users/mghavami/.local/easybuild
module-naming-scheme (E) = CategorizedModuleNamingScheme
prefix             (E) = /home/users/mghavami/.local/easybuild
repositorypath     (D) = /home/users/mghavami/.local/easybuild/ebfiles_repo
robot-paths        (D) = /opt/apps/resif/aion/2020b/epyc/software/EasyBuild/4.5.4/easybuild/easyconfi
sourcepath         (D) = /home/users/mghavami/.local/easybuild/sources
0 [mghavami@aion-0104 eessi](6910572 1N/1T/1CN)$

```

Figure 4: Answers to the question

- What is the current version of EasyBuild available on Aion?
4.5.4
- What are the options with non-default values in the EasyBuild configuration? `buildpat`, `prefix`, `module-naming-scheme`
- What is the installation path used by EasyBuild?
`/home/users/mghavami/.local/easybuild`

5 Question 6: Installing the latest EasyBuild

Answer to these questions in your report:

- What is the command line to install the latest version of EasyBuild with EasyBuild?

```

1      module purge
2      module load lang/Python/3.8.6-GCCcore-10.2.0
3      eb --install-latest-eb-release
4      module load tools/EasyBuild/5.0.0
5      eb --version

```

- What is the version of the newly installed easybuild?//

```
0 [mghavami@aion-0185 ~](6948342 1N/1T/1CN)$ eb --version
This is EasyBuild 5.0.0 (framework: 5.0.0, easyblocks: 5.0.0) on host aion-0185.
```

Figure 5: Answers to the question

6 Question 7: Installation of HPL

Write the answer in your report:

- What is the full path of the EasyConfig file that you used to build HPL?
/home/users/mghavami/.local/easybuild/software/EasyBuild/5.0.0/easybuild/easyconfigs/h/HPL/HPL-2.3
- What is the version of HPL built with this EasyConfig file?
2.3
- Add the log file generated by EasyBuild to the hpl directory of your GitHub Classroom repository
Done!
- In your opinion, is it easier to install HPL manually or with EasyBuild? Why?
Of course it's much easier with EasyBuild since it needs just one line code. Also, it is easy to find and install the desired version.

7 Question 8: Installation of OSU Micro Benchmarks 7.5

- What is the EasyBuild command line to install the latest OSU Micro Benchmarks on Aion?

```
1 eb --try-software=OSU-Micro-Benchmarks,7.5 --try-toolchain=gompi,2020b
```

- Add the generated EasyConfig file to the omb directory of your GitHub Classroom repository.
Done!
- Add the EasyBuild log file to the omb directory of your GitHub Classroom repository.
Done!

8 Question 9: EasyConfig file for MDI 1.4.29

Add the following file to the mdi directory of your GitHub Classroom repository:

- Your new EasyConfig file for MDI 1.4.29
- The EasyBuild log file of the installation

Why is it needed to update the checksum? What is the purpose of this checksum?

9 Question 10: Exploring the EESSI environment

Check your environment to answer the following questions:

- What is the processor architecture detected by EESSI? You can use `env | grep EESSI` to see the environment variables related to EESSI.
=x86₆4
- What is the path to the modulefiles provided by EESSI? Explain how you found the answer.
/cvmfs/software.eessi.io/versions/2023.06/software/linux/x86₆4/amd/zen2/modules/all

- What is the name of the OpenFOAM 11 module provided by EESSI? Explain how you found the answer. OpenFOAM/11-foss-2023a, by using `module avail openfoam`
- What are the toolchains available with EESSI? Explain how you found the answer. `foss`, `gfbf`, `gomp`, by opening the directory

Take a screenshot of your terminal output and add it to your report.

```

[mghavami@aion-0015 motorBike-INPUT](6956248 1N/1T/1CN)$ env | grep EESSI
EESSI_PREFIX=/cvmfs/software.eessi.io/versions/2023.06
EESSI_OS_TYPE=linux
EESSI_SITE_SOFTWARE_PATH=/cvmfs/software.eessi.io/host_injections/2023.06/software/linux/x86_64/amd/zen2
EESSI_SITE_MODULEPATH=/cvmfs/software.eessi.io/host_injections/2023.06/software/linux/x86_64/amd/zen2/modules/all
EESSI_CVMFS_REPO=/cvmfs/software.eessi.io
EESSI_EPREFIX=/cvmfs/software.eessi.io/versions/2023.06/compat/linux/x86_64
EESSI_VERSION=2023.06
EESSI_CPU_FAMILY=x86_64
EESSI_ACCEL_SUBDIR=
LOADEDMODULES=env/release/default:EESSI/2023.06
EESSI_SOFTWARE_PATH=/cvmfs/software.eessi.io/versions/2023.06/software/linux/x86_64/amd/zen2
EESSI_MODULEPATH=/cvmfs/software.eessi.io/versions/2023.06/software/linux/x86_64/amd/zen2/modules/all
EESSI_ARCHDETECT_OPTIONS=x86_64/amd/zen2:x86_64/generic
LMFILES_=/opt/apps/easybuild/environment/modules/env/release/default.lua:/cvmfs/software.eessi.io/init/modules/EESSI/2023.06.lua
EESSI_SOFTWARE_SUBDIR=x86_64/amd/zen2
[mghavami@aion-0015 motorBike-INPUT](6956248 1N/1T/1CN)$ module avail openfoam

----- /cvmfs/software.eessi.io/versions/2023.06/software/linux/x86_64/amd/zen2/modules/all -----
OpenFOAM/v2312-foss-2023a   OpenFOAM/v2406-foss-2023a   OpenFOAM/10-foss-2023a   OpenFOAM/11-foss-2023a (D)
----- /opt/apps/resif/aion/2020b/epyc/modules/all -----

cae/OpenFOAM/8-foss-2020b

Where:
D: Default Module

If the avail list is too long consider trying:

[mghavami@aion-0015 eessi](6956248 1N/1T/1CN)$ ls /cvmfs/software.eessi.io/versions/2023.06/software/linux/x86_64/amd/zen2/modules/toolchain/
foss  gfbf  gomp

```

Figure 6: Answers to the question

10 Question 11: Using OpenFOAM provided by EESSI

Add the following files to the `eessi` directory of your GitHub Classroom repository:

- Your modified `run_OpenFOAM_parallel.sh` script
Done!
- Your execution log file `SLURM_motorbike_XXX.log`
Done!